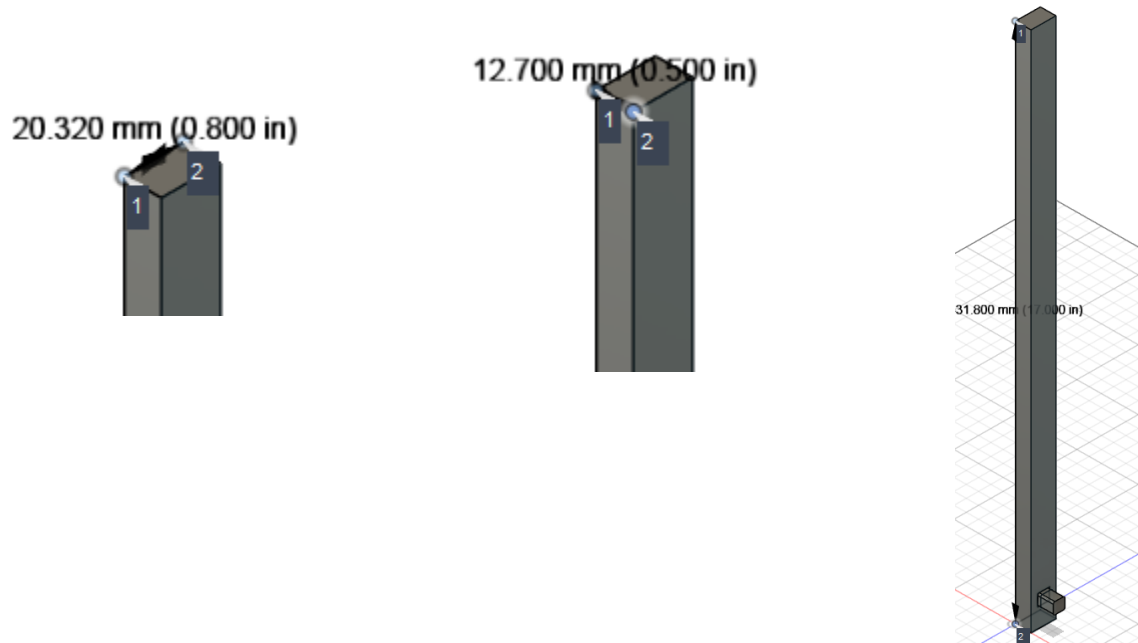


MAE 3270 Final Project Part II

Performed by Frederick Wentzien working with Justice Rose

Performed on 12/10/25

Cad Model



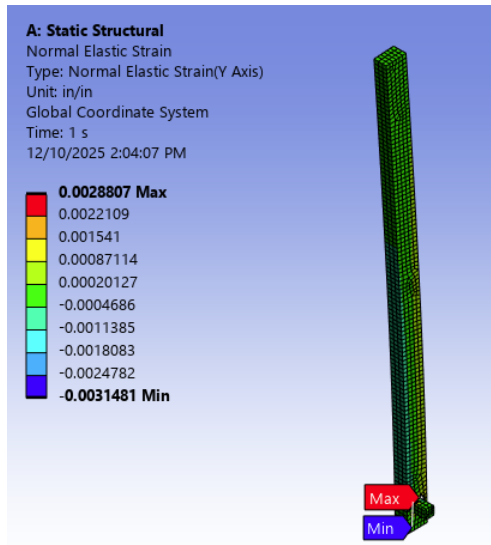
Material Used

The material used for this design is aluminum 747 T7651, which is an aluminum alloy with high strength and fracture toughness. The material properties relevant to this investigation are a Young's modulus of 10.5E6 psi, a Poisson's ratio of 0.33, a yield strength of 65.2 ksi, a fracture toughness of 45.8 ksi·in^{0.5}, and a fatigue strength of 46 ksi. What this means is that this has a very high fracture toughness and can resist fracture well. As well, it still has a high strength and fatigue resistance.

Loads and Boundary Conditions

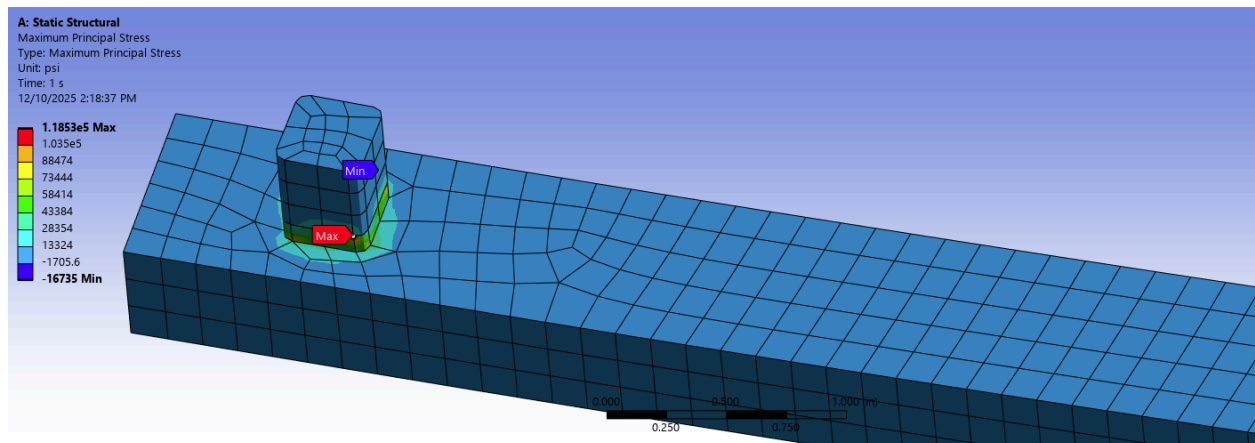
The loads were applied to my model by utilizing the load feature on Ansys and specifying a 37.5 lb force in the z direction, such that it applied a 600 in-lb torque. The boundary conditions were applied by specifying that the top 4 inches of the drive were stationary and could not move in the x, y, or z direction.

Normal Strain Contours



The normal strain contours in the y direction, or the direction of the strain gauge, show a maximum strain of 0.0028807 strains.

Maximum Principal Stresses



Results of FEM Calculations

Maximum Normal Stress	53741 psi
Load Point Deflection	0.393 Inches
Strain at Strain Gauge Deflection	1.05 Microstrains

*Please see Appendix A for comparison to hand calculations

Torque Wrench Sensitivity

Using the strain gauge output we found a torque wrench sensitivity of around 1.1 mv/V.

Strain Gauge

The strain gauge configuration that was selected was a half-bridge setup. The dimensions of the strain gauge selected were 0.24x0.25 inches and the type is linear.

Appendix

A)Hand Calculations

%Material Properties

name = 'Aluminum 7475 T7651'; % material name

M = 600; % max torque (in-lbf)

L = 16; % length from drive to where load applied (inches)

h = 0.8; % width

b = 0.5; % thickness

c = 1.0; % distance from center of drive to center of strain gauge

E = 10.E6; % Young's modulus (psi)

nu = 0.29; % Poisson's ratio

su = 65.E3; % tensile strength use yield or ultimate depending on material (psi)

KIC = 46.E3; % fracture toughness (psi sqrt(in))

sfatigue = 40.e3; % fatigue strength from Granta for 10⁶ cycles

I = (b*(h³))/12;

P = M/L;

cr = 0.04; %Crack length (inches)

%Stress and deflection analysis Using Beam Theory

Umax = (P*(L³))/(3*E*I);

SNmax = (M*(h/2))/I;

%Safety Factor Results

Xo = (su*I)/((h/2)*M); %4

Xk = (KIC/M)*((b*h²)/6)*(1/(1.12*sqrt(pi)*sqrt(cr))); %2

Xs = ((b*h²)/6)*(sfatigue/M); %1.5

%Strain Gauge

e = (P*(L-c))*(h/2)*(1/(E*I));

output = 1000*e;

Outputs:

X0: 5.78

Xk: 10.3

Xs: 3.55

Strain Gauge Output: 1.1 at Half Bridge